SCNet: Lightweight Spatial–Channel Attention Network for Remote Sensing Change Captioning

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***Abstract*—Change captioning has become essential for accurately describing changes in multitemporal remote sensing data, providing an intuitive way to monitor Earth's dynamics through natural language. However, existing change captioning methods face two key challenges. First, the use of multistage feature fusion strategies aims to achieve better change detection results by combining information from multiple stages. This approach leads to high computational demands, as it requires intensive processing across different feature levels. Second, previous methods mainly focus on feature extraction and fusion along the image spatial dimensions, overlooking the importance of channel-wise information. This leads to insufficient semantic extraction, resulting in inadequate detail in object descriptions and limiting the ability to fully capture the changes. To solve these challenges, we propose a spatial–channel attention encoder (SCE) based on the transformer model for remote sensing change captioning, which is named SCNet. In particular, SCNet integrates an SCE, a difference-guided fusion module, and a caption decoder. Compared to typical models that require multistage fusion in the transformer encoder and fusion module. By jointly modeling spatial and channel information in SCE, our approach significantly enhances the model's ability to extract semantic information from objects in multitemporal remote sensing images. Extensive experiments validate the effectiveness of SCNet, achieving CIDEr scores of 140.23% on the LEVIR-CC dataset and 97.74% on the DUBAI-CC dataset, surpassing current state-of-the-art (SOTA) methods. The code and pretrained models will be available at SCNet**

***Index Terms*—Change captioning, deep learning, image captioning, remote sensing.**

I. Introduction

With the rapid development of geoscience and remote sensing technologies, the collection and analysis of Earth observation data have become increasingly sophisticated. Leveraging advances in artificial intelligence (AI) and natural language processing (NLP), vision-language models have recently gained traction in the remote sensing community, enabling richer human–AI interaction and multidimensional data interpretation [1], [2], [3], [4], [5], [6]. These models integrate visual and textual information, supporting a range of tasks that extend beyond single-modality processing. Among them, as shown in Fig. 1, four core tasks—image captioning [7], [8], [9], text-to-image generation [10], [11], visual question answering (VQA) [12], [13], [14], [15], and image–text retrieval [16], [17], [18]—have laid the foundation for further advancements in vision-language understanding in the geospatial domain.

Meanwhile, the widespread availability of multitemporal satellite imagery has spurred research in Earth surface monitoring and dynamic change detection [19], [20], [21], [22], [23]. Traditional change detection methods focus on generating pixel-wise binary masks that classify each pixel as "changed" or "unchanged" [24], [25]. However, such outputs lack semantic interpretability and are not intuitive for nontechnical domain users. To bridge this gap, researchers have introduced the task of remote sensing image change captioning (RSICC) [26], [27], which aims to describe observed land cover changes between bi-temporal images using natural language.

RSICC facilitates an intuitive, high-level understanding of changes, as illustrated in the bottom line of Fig. 1, which is particularly valuable for urban development [28], disaster response, and environmental monitoring [29]. For example, instead of only providing a binary mask, a model can produce a textual description such as "two new residential buildings have been constructed in the southern area" offering clearer insights for decision-makers [30]. Despite its potential, RSICC

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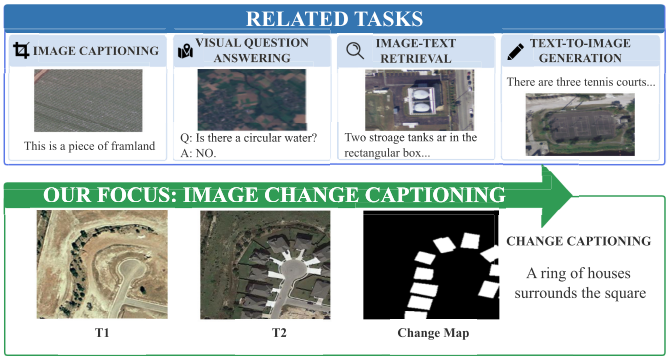


Fig. 1. Overview of remote sensing vision-language tasks. While standard tasks such as captioning, VQA, and retrieval focus on interpreting static information, remote sensing change captioning requires understanding dynamic relationships and describing changes between bi-temporal image pairs.

remains challenging due to the complexity of remote sensing imagery—featuring high interclass similarity, complex backgrounds, and significant variation in object scales and appearances.

Earlier approaches to RSICC used CNN-based backbones with RNN or LSTM decoders [26], [27]. More recent methods incorporate transformer architectures with multistage fusion of bi-temporal features, such as RSICCFormer [26], Chg2Cap [31], and PSNet [32]. While these methods enhance captioning performance, their reliance on multistage fusion increases model complexity and often overlooks object-level semantics. This limits their capacity to interpret subtle or small-scale changes accurately.

To address these limitations, we propose a novel method named **SCNet**, which introduces a spatial–channel attention encoder (SCE) based on self-attention mechanisms. The SCE sequentially captures spatial and channel dependencies, effectively modeling both the relationships between image regions and the semantic attributes of objects. To improve local feature modeling, we incorporate depth-wise convolutions into the transformer encoder. Furthermore, we share encoder weights between bi-temporal branches to enhance the efficiency. Instead of complex multistage fusion, our approach adopts a lightweight strategy that concatenates the refined bi-temporal features along the channel dimension, fuses them through a residual block, and feeds the result into a transformer decoder for caption generation.

We evaluate our method on two publicly available RSICC benchmarks: LEVIR-CC [26] and DUBAI-CCD [27]. Experimental results show that our method achieves CIDEr scores of **140.23%** and **97.74%** on LEVIR-CC and DUBAI-CCD, respectively. Compared with existing approaches, SCNet achieves better accuracy with fewer parameters, validating the effectiveness of our spatial–channel modeling and simplified fusion design.

The main contributions of this article are summarized as follows.

1) *Bi-Temporal Spatial–Channel Modeling*: To address the limitation of insufficient semantic extraction in existing methods, we propose a Siamese SCE tailored for the bi-temporal nature of RSICC. Unlike generic attention mechanisms, SCE explicitly decouples and models "where changes occurred" (spatial dependencies) and "what semantic attributes changed" (channel dependencies). By sharing weights between temporal branches, this design ensures robust feature alignment and captures subtle object-level transformations effectively.
2) *Lightweight "Encoder-Heavy, Fusion-Light" Architecture*: Challenging the prevailing trend of computationally expensive multistage fusion networks (e.g., Chg2Cap and RSICCFormer), we propose a streamlined difference-guided fusion strategy. We demonstrate that by refining features intensively in the encoder, the fusion stage can be simplified to a lightweight residual block without performance loss. Combined with depth-wise convolutions, SCNet achieves a significant reduction in model complexity, containing only 7% of the parameters of state-of-the-art (SOTA) heavy models while maintaining high descriptive accuracy.
3) *SOTA Performance With Efficiency*: We conduct extensive experiments on the LEVIR-CC and DUBAI-CCD benchmarks. SCNet achieves new SOTA CIDEr scores of 140.23% and 97.74%. Crucially, our results prove that a shallow, carefully designed attention network can outperform deep, redundant transformer architectures, offering a superior tradeoff between captioning quality and computational efficiency for remote sensing applications.

The rest of this article is organized as follows. Section II introduces the related works of remote sensing change captioning. Section III describes the proposed method in detail. Section IV presents the information on LEVIR-CC and DUBAI-CCD datasets used in this study and the experimental results. Conclusions and other discussions are summarized in Section V.

II. Related Works

A. Deep Learning in Remote Sensing Analysis

With the rapid advancement of deep learning, intelligent interpretation of remote sensing imagery has achieved remarkable success across various tasks, ranging from fine-grained object detection and tracking [33], [34], [35], [36] to high-level semantic understanding. Building on these foundational works, RSICC [37], [38], [39] has emerged as a vision-language task designed to generate coherent descriptions of land cover changes between bi-temporal images. This task is critical for urban planning and disaster monitoring [40], requiring models to capture both dynamic temporal relationships and static semantic attributes.

B. Evolution of RSICC Models

Early approaches to RSICC primarily utilized CNN-RNN architectures. Hoxha et al. [27] pioneered this field by introducing the LEVIR-CC and DUBAI-CCD datasets and employing LSTMs for caption generation. However, these RNN-based methods often struggled to capture long-range dependencies and fine-grained visual details. To address these

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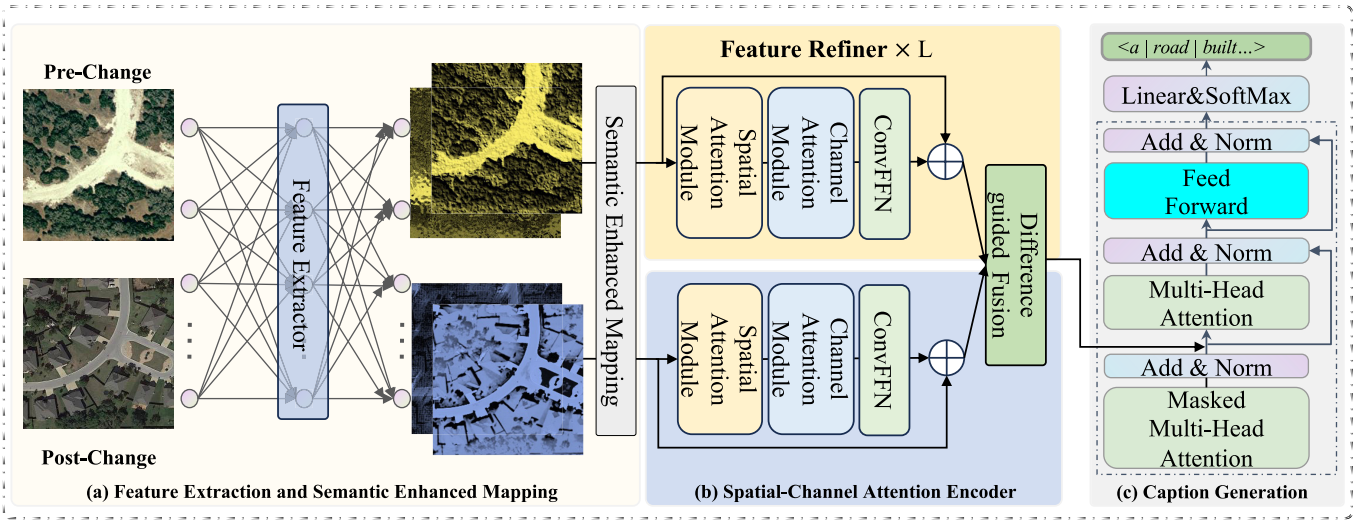


Fig. 2. Overview of the SCNet framework. (a) Feature extraction and SEM: utilize a pretrained ResNet-101 and depth-wise convolutions to extract and refine local feature details. (b) SCE: adopt a siamese weight-sharing structure to sequentially model spatial dependencies ("where changes occurred") and channel attributes ("what objects changed"). (c) Caption generation: fuse the bi-temporal features via a lightweight difference-guided block and generate the final description using a transformer decoder.

limitations, transformer-based architectures with multistage fusion strategies became the dominant paradigm. Liu et al. [26] proposed RSICCFormer, which utilizes a difference encoding module and multistage bitemporal fusion. Similarly, PSNet [32] leverages a progressive multiscale fusion strategy, and Chg2Cap [31] employs a dual self-attention encoder with feature fusion. However, a major limitation of these methods is their reliance on computationally expensive fusion mechanisms and deep architectures, for example, Chg2Cap requires 286 M parameters. Such "fusion-heavy" designs significantly increase model complexity and latency, making them less suitable for resource-constrained applications.

C. Recent Advances and Limitations

More recently, research has shifted toward leveraging large language models (LLMs) and novel prompt learning paradigms [41]. For instance, Prompt-CC [42] decouples change detection and identification using CLIP-based learnable prompts, while RSCaMa [43] explores state-space models for joint spatio-temporal modeling. Other works have integrated pixel-level auxiliary tasks [44], [45], [46] or unified change detection and captioning into a single framework [47], [48], [49]. Despite these advancements, most existing methods focus on improving accuracy through increasingly complex architectures or massive pretraining. They often overlook the efficiency-accuracy tradeoff and lack explicit mechanisms to decouple spatial ("where") and channel ("what") semantics efficiently. In contrast, our proposed SCNet introduces a lightweight SCE to address these gaps, achieving SOTA performance with significantly fewer parameters.

III. Methodology

A. Overview

Our proposed SCNet method comprises four main components: a feature extraction module, an SCE, a difference-guided fusion module, and a standard transformer-based caption decoder. The model takes as input a bi-temporal image pair along with the corresponding change caption. The overall flow of the model architecture is illustrated in Fig. 2.

First, the feature extraction module utilizes a pretrained ResNet-101 [50] backbone to extract feature maps from the input image pair. These feature maps are then processed by the SCE, which jointly models spatial and channel dependencies. The spatial attention mechanism captures changes across different regions of the image, while the channel attention mechanism extracts rich semantic information about objects within the image.

After refinement in the encoder, the two feature maps are passed into the fusion module, where they are concatenated along the channel dimension to form a unified visual embedding. This visual embedding is subsequently fed into the transformer decoder, which generates change captions for the bi-temporal image pair.

B. Image Pair Feature Extraction

Given an image pair (I_1, I_2), we typically use a pretrained CNNs to extract features, where $I_i \in \mathbb{R}^{C\times H\times W}$, $i \in (1, 2)$, and C, H, and W represent the channels, height, and width of the image, respectively. In this article, we use ResNet-101 [50] as our backbone to extract feature maps (X_1, X_2). Additionally, we remove the dropout layer and classification layer. Since the channel dimension of the feature maps extracted by CNNs is 2048, to reduce the complexity of the model, we need to decrease the channel dimension. However, if we simply use a convolution layer on the feature maps, it will result in information loss, preventing the model from capturing the semantic information of specific changes in small objects optimally. Therefore, the model finds it more difficult to distinguish between different objects, leading to poorer quality captions when the image pair contains only minor changes.

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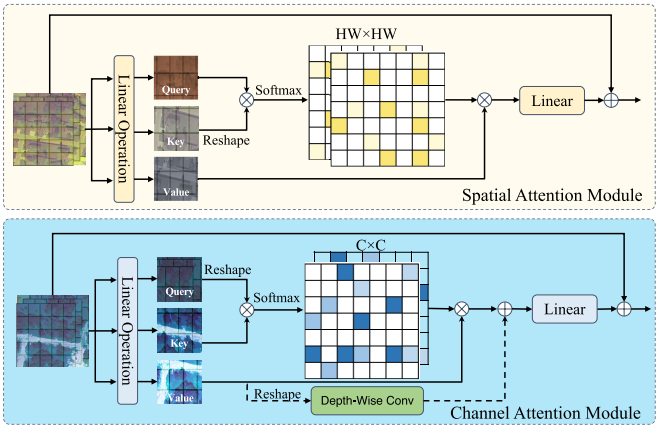


Fig. 3. Illustration of the two attention mechanisms used in SCNet. (Top) SAM: compute global spatial correlations to precisely locate change regions ("where changes occurred"). (Bottom) CAM: model interchannel dependencies to capture rich semantic attributes ("what objects changed"). Crucially, CAM incorporates depth-wise convolutions to preserve local contextual details, ensuring that the model performs well on small targets while remaining lightweight.

To learn as precise semantic information as possible, we propose a semantic-enhanced mapping (SEM) module. Compared to simply mapping features extracted by CNNs to a feature space through one convolutional layer, our module can learn richer fine-grained information from the images. Specifically, our module first incorporates positional encoding on the features extracted by the backbone (X_1, X_2), for the input $X_i \in \mathbb{R}^{C^o \times H \times W}$

$$X_i' = X_i + X_{pos} \tag{1}$$

where $i = 1$ represents the input before the change and $i = 2$ represents the input after the change. $X_{pos} \in \mathbb{R}^{C^o \times H \times W}$ is a learnable 2-D position embedding.

Then, we feed the features extracted by the backbone (X_1', X_2') into the SEM module, which consists of two standard convolutional layers with batch normalization (BN) and rectified linear unit (ReLU) function, and one Depthwise convolution layer. The process is given as follows:

$$X_i'' = \text{Conv}_2\left(\text{Conv}_1\left(X_i'\right)\right) \tag{2}$$

$$F_i = \text{DWConv}\left(X_i''\right) \tag{3}$$

where $\text{Conv}_i(\cdot)(i = 1, 2)$ are two standard 1×1 convolutional layers with BN and ReLU. $\text{DWConv}(\cdot)$ represents the 3×3 depth-wise convolution layer. Through the SEM module, we obtain a pair of locality-enhanced feature maps, which enhances the model's ability to capture and model small objects.

C. Spatial–Channel Attention Encoder

The SCE is composed of several stacked transformer blocks, each consisting of three parts: spatial attention module (SAM), channel attention module (CAM), and convolutional feed-forward network (ConvFFN), as shown in Fig. 3.

1) Spatial Attention Module: In the SAM, we utilize the self-attention mechanism [51], which is widely used in transformer-based methods, to capture the spatial information of the feature map. Given the input $F_i \in \mathbb{R}^{C \times H \times W}$, we first reshape it to $\mathbb{R}^{HW \times C}$ and then generate query Q, key K, and value V through three linear layers. This process can be represented as follows:

$$Q = F_i W_Q, \quad K = F_i W_K, \; V = F_i W_V \tag{4}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{C \times C}$ are linear layer learnable parameters. Subsequently, we divide Q, K, and V into multiple heads h, with each head having a dimension $\mathbb{R}^{C \times (C/h)}$. Calculate attention separately in each head, and then, concatenate them together to obtain our spatial feature maps. The entire process is as follows:

$$\text{Head}_i(X) = \text{Softmax}\left(\frac{Q_i K_i^{\mathrm{T}}}{\sqrt{d_k}}\right) V_i \tag{5}$$

$$F_i' = \text{Concat}\left(\text{Head}_1(F_i), \ldots, \text{Head}_h(F_i)\right) W \tag{6}$$

where $\text{Head}_i(F_i)$ is the ith head dot-product attention, $W \in \mathbb{R}^{C \times C}$ is the learnable weight matrix to fuse features, and F_i' is the output feature map.

2) Channel Attention Module: Most existing RSICC methods primarily focus on spatial-based attention mechanisms. Although some recent approaches [43] introduce channel or group-wise interactions, the exploitation of channel-wise dependencies in RSICC remains relatively limited or implicit. As shown in previous works [9], [52], different channels often contain distinct semantic information and attributes. In tasks like change captioning, which involve two input images simultaneously, it is crucial to capture more precise semantic information and attributes. Therefore, CAM is introduced to model the relationship of input feature maps along the channel dimension. Specifically, it generates a global attention map by computing cross-covariance across channels to encode feature maps' semantic information. Here, we also employ a self-attention mechanism similar to SAM, with the only difference being that in CAM, this attention calculation is conducted along the channel dimension. Additionally, to enhance the module's ability to understand local contextual features, we directly incorporate depth-wise convolution on the value V.

Given the input features $F_i' \in \mathbb{R}^{HW \times C}$, our CAM first generates Q, K, and V through a linear layer. Next, Q, K, and V are reshaped to their corresponding size $\mathbb{R}^{HW \times C}$ so that channel-wise feature maps can be generated through the dot-product operation of Q and K. Meanwhile, we apply depth-wise convolution to V to extract local contextual relationships. The overall process can be summarized as follows:

$$\left[Q_i'; K_i'; V_i'\right] = F_i' W + b \tag{7}$$

$$\text{Attention}\left(Q_i', K_i', V_i'\right) = V_i' \cdot \text{Softmax}\left(\frac{\left(Q_i'\right)^T K_i'}{\alpha}\right) + \text{DWConv}\left(V_i'\right) \tag{8}$$

$$F_i'' = \text{Attention}\left(Q_i', K_i', V_i'\right) + F_i' \tag{9}$$

where $W \in \mathbb{R}^{C \times 3C}$ and $b \in \mathbb{R}^{3C}$ represent the projection weight and bias. α is a learnable parameter to adjust the dot-product,

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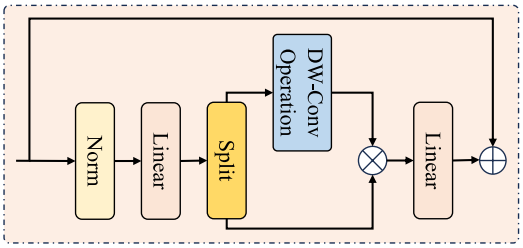


Fig. 4. ConvFFN Module. This module enhances local information modeling by incorporating a depth-wise convolutional layer within the standard feed-forward network structure. This modification allows the network to capture local spatial context more effectively, which is crucial for processing image features compared to traditional linear FFNs in text-based transformers.

which is initialized to 1. DWConv(·) represents the 3×3 depth-wise convolution layer with BN and GELU nonlinear function [53]. F_i and F_i' are the input and output features.

3) Convolutional Feed-Forward Network: In the standard transformer model, the FFN typically consists of two linear layers and a nonlinear activation function. Since the original transformer architecture was designed for modeling text information, when applied to the computer vision domain, it ignores local information in images. Therefore, in many previous works, DWconv is introduced into the FFN to enhance its capability to model local information

$$Z_i = \mathrm{DWConv}\left(\sigma\left(F_i'' W_1 + b_1\right)\right) W_2 + b_2 \tag{10}$$

where $W_1 \in \mathbb{R}^{C\times 4C}$ and $W_2 \in \mathbb{R}^{4C\times C}$ and $b_1 \in \mathbb{R}^{4C}$ and $b_2 \in \mathbb{R}^{C}$ represent the linear projection weights and biases. $\sigma(\cdot)$ is a nonlinear activation function. DWConv(·) represents the 3×3 depth-wise convolution layer. F_i'' and Z_i are the input and output feature maps. However, adding DWConv significantly increases the parameters of the module. In order to improve the computational efficiency of the module, we first reshape and then divide the feature after the activation function into two features $L_i^1, L_i^2 \in \mathbb{R}^{2C\times H\times W}$ along the channel dimension. Then, we multiply these two features and thereby reduce the parameter count of the second linear layer. ConvMLP is illustrated in Fig. 4 and formulated as

$$\left[L_i^1; L_i^2\right] = \sigma\left(F_i'' W_1 + b_1\right)\right) \tag{11}$$

$$\hat{L}_i^1 = \mathrm{DWConv}\left(L_i^1\right) \tag{12}$$

$$Z_i = \left(\hat{L}_i^1 \odot L_i^2\right) W_2' + b_2' \tag{13}$$

where $W_2' \in \mathbb{R}^{2C\times C}$ and $b_2' \in \mathbb{R}^{C}$ represent the downsample projection weight and bias. $\hat{L}_i^1 \in \mathbb{R}^{2C\times H\times W}$ is the feature obtained after 3×3 depth-wise convolution layer.

D. Difference-Guided Fusion

After obtaining the feature maps refined by the spatial–channel attention encoder for bi-temporal image pairs, we design a difference-guided fusion block to integrate information from the two feature maps and guide the caption decoder. The overall idea of leveraging feature differences is conceptually related to Change2Captions (Chg2Cap) [31]; however, our implementation differs in both feature interaction and refinement strategies. Specifically, we compute the cosine similarity between bi-temporal feature maps and inject it into each feature map to emphasize change-related responses. The enhanced features are then concatenated and passed through a compact convolutional refinement module with residual connections for dimensionality reduction and feature enhancement, enabling efficient integration with the proposed spatial–channel attention encoder. The fusion process is formulated as

$$Sim = \mathrm{Cos}\left(Z_1, Z_2\right) \tag{14}$$

$$F_f = \left[Z_1 + Sim; Z_2 + Sim\right] \tag{15}$$

$$F_f' = \mathrm{Conv}_3\left(\mathrm{Conv}_2\left(\mathrm{Conv}_1\left(F_f\right)\right)\right) + F_f \tag{16}$$

where $\mathrm{Cos}(\cdot,\cdot)$ can calculate the cosine similarity between two tensors. $\mathrm{Conv}_i(\cdot)(i = 1, 2, 3)$ are three convolutional layers with BN and ReLU; their kernel sizes are 1×1, 3×3, and 1×1. After obtaining the fused feature $F_f' \in \mathbb{R}^{C\times H\times W}$, we reshape it and feed it into the decoder of the model to guide the decoder in generating text describing the changes.

E. Caption Generation

To generate text describing changes in remote sensing image pairs, we use a widely adopted transformer-based decoder. It consists of a series of stacked transformer decoder blocks, where each transformer decoder block comprises a masked multiheaded self-attention mechanism, an encoder–decoder cross-attention mechanism, and a feed-forward network. The calculation within a transformer decoder block can be represented as follows:

$$X_l' = \mathrm{LN}\left(X_{l-1} + \text{M-MSA}\left(X_{l-1}\right)\right) \tag{17}$$

$$X_l'' = \mathrm{LN}\left(X_l' + \mathrm{CA}\left(X_l', F_f', F_f'\right)\right) \tag{18}$$

$$X_l = \mathrm{LN}\left(X_l'' + \mathrm{MLP}\left(X_l''\right)\right) \tag{19}$$

where X_{l-1} and X_l denote the input and output of the lth transformer decoder block, respectively, and M-MSA, CA, MLP, and LN denote the masked multiheaded self-attention, encoder–decoder cross-attention, feed-forward network, and layer normalization, respectively.

IV. Experiments

A. Dataset Descriptions

We adopt two benchmark datasets to evaluate the performance of the proposed method on RSICC tasks.

1) LEVIR-CC Dataset: The images in this dataset [26] are mainly collected from the LEVIR-CD dataset [54], which consists of images from 20 regions in Texas, USA. Each image has dimensions of 1024×1024 pixels, with a spatial resolution of 0.5 m. The time span for each change image pair ranges from 5 to 15 years. LEVIR-CC comprises a total of 10 077 pairs of images, each with dimensions of 256×256 pixels. Additionally, each image pair is accompanied by five corresponding texts describing whether changes have occurred. The maximum length of the text is 39, and the average length of the text is 7.99. In total, LEVIR-CC contains 5038 pairs of images with changes and 5039 pairs without changes. We use a predefined partition of the dataset to validate the effectiveness of the model, with 6815 image pairs used for training, 1333

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TABLE I
COMPARISON WITH THE STATE OF THE ART ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
Capt-Rep-Diff [37]	72.90	61.98	53.62	47.41	34.47	65.64	110.57	64.52
Capt-Att [37]	77.74	67.40	59.24	53.15	36.58	69.73	121.22	70.17
Capt-Dual-Att [37]	79.51	70.57	63.23	57.46	36.56	70.69	124.42	72.28
DUDA [37]	81.44	72.22	64.24	57.79	37.15	71.04	124.32	72.58
MCCFormer-S [55]	79.90	70.26	62.68	56.68	36.17	69.46	120.39	70.68
MCCFormer-D [55]	80.42	70.87	62.86	56.38	37.29	70.32	124.44	72.11
RSICCFormer [26]	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.65
PSNet [32]	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
Prompt-CC [42]	83.66	75.73	69.10	63.54	38.83	73.72	136.44	78.13
Chg2Cap [31]	**86.14**	78.08	70.66	64.39	40.03	75.12	136.61	79.04
RSCaMa [43]	85.79	77.99	71.04	65.24	39.91	75.24	136.56	79.24
Sparse Focus [56]	84.56	75.87	68.64	62.87	39.93	74.69	137.05	78.64
DiffusionRSCC [57]	-	-	-	60.90	37.80	71.50	125.60	73.95
Mask Approx Net [30]	85.90	77.12	70.72	64.32	39.91	**75.67**	137.71	79.40
SEN [58]	85.10	77.05	70.01	64.09	39.59	74.57	136.02	78.57
SCNet(ours)	**86.14**	**78.19**	**71.44**	**65.82**	**40.51**	75.37	**140.23**	**80.48**

image pairs for validation, and the remaining 1929 image pairs for testing.

2) Dubai-CCD Dataset: This dataset [27] describes the urban changes in Dubai between May 19, 2000, and June 16, 2010, and includes a total of 500 bi-temporal image pairs, each with dimensions of 50 × 50 pixels. Each image pair is accompanied by five related change description captions, with a maximum length of 23 words and an average length of 7.35 words. We use the predivided portions of the dataset, with 300 image pairs serving as the training set, 50 image pairs as the validation set, and the remaining 150 image pairs as the test set to evaluate the model's effectiveness.

3) WHU-CDC Dataset: The WHU-CDC dataset is constructed based on the WHU-CD dataset [59] and consists of ultrahigh-resolution image pairs (0.075-m spatial resolution) with original dimensions of 32 507 × 15 354 pixels. The dataset captures post-earthquake urban changes in the Christchurch area of New Zealand, particularly focusing on building construction following the 6.3-magnitude earthquake in February 2011. Building upon this source, WHU-CDC [60] contains 7434 paired images, which are uniformly resized to 256 × 256 pixels for model training and evaluation. Each image pair is annotated with five change descriptions, resulting in a total of 37 170 sentences. The textual annotations vary in length from 3 to 24 words and comprise a vocabulary of 327 distinct terms.

B. Experimental Settings and Implementation Details

In our experiments, we set the maximum number of iterations to 50 epochs for both LEVIR-CC dataset and DUBAI-CCD dataset. The Adam optimizer with a learning rate of $1e^{-4}$ is employed during the training phase. We use a batch size of 32 for the LEVIR-CC dataset and a batch size of 8 for the DUBAI-CCD dataset. We use a pretrained ResNet-101 to extract feature maps from the image pairs. The feature dimensions in both the transformer encoder and transformer decoder are set to 512, the hidden dimension in the feed-forward layer is set to 2048, and the number of self-attention heads is 8. Through experimental validation, we set the number of layers for transformer encoder to 3 and the number of layers for transformer decoder to 1. Since ResNet-101 is pretrained on natural image datasets, to better adapt it to the RSICC task, we first fine-tune ResNet-101 on the current datasets and then use the fine-tuned model as our backbone. The experiment is conducted on an Nvidia RTX A6000 GPU.

Following the standard evaluation metrics in previous works [26], [31], BLEU-N (where N = 1,2,3,4) [61], ROUGE-L [62], METEOR [63], CIDEr-D [64], and S_m^* [65] as an overall metric, which is simply the average of the BLEU-4, METEOR, ROUGE-L, and CIDEr scores are adopted to evaluate our proposed method and compared with other SOTA methods.

C. Performance Comparison to the State of the Art

In this section, we compare the proposed SCNet with SOTA methods on three benchmark datasets: LEVIR-CC, DUBAI-CCD, and WHU-CDC. The baselines include representative models such as RSICCFormer [26], Chg2Cap [31], RSCaMa [43], Sparse Focus [56], Mask Approx Net [30], and SEN [58].

1) *LEVIR-CC Dataset (Table I):* The results demonstrate the comprehensive superiority of SCNet, which achieves the highest scores across **all** evaluation metrics. Notably, our method attains a CIDEr-D score of **140.23%**, surpassing the runner-up (Sparse Focus) by a significant margin of 3.18%. Improvements in precision metrics (e.g., 65.82% in BLEU-4 and 75.37% in ROUGE-L) further validate the model's ability to produce precise descriptions in complex scenes.
2) *DUBAI-CCD Dataset (Table II):* On the challenging desert-urban DUBAI dataset, SCNet demonstrates strong robustness, achieving SOTA performance in most key metrics. Most significantly, it achieves a **CIDEr-D score of 97.74%**, outperforming the strong baseline Chg2Cap by **5.25%**. The method also leads in METEOR (29.62%) and ROUGE-L (59.06%), proving its effectiveness in handling distinct geographical changes.

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TABLE II
COMPARISON WITH THE STATE OF THE ART ON THE DUBAI-CCD DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	$S_m^\star$
CC-RNN-sub [27]	67.19	52.23	40.00	28.54	25.51	51.78	69.73	43.89
CC-SVM-sub [27]	70.71	57.58	46.10	35.50	27.60	56.61	82.96	50.67
DUDA [37]	58.82	43.59	33.63	25.39	22.05	48.34	62.78	39.64
MCCFormer-S [55]	52.97	37.02	27.62	22.57	18.64	43.29	53.81	34.58
MCCFormer-D [55]	64.65	50.45	39.36	29.48	25.09	51.27	66.51	43.09
RSICCFormer [26]	67.92	53.61	41.37	31.28	25.41	51.96	66.54	43.80
ICT-Net [67]	69.38	57.03	46.50	36.17	26.78	57.31	92.97	53.31
Prompt-CC [42]	70.03	58.41	49.44	40.32	26.48	55.82	85.44	52.01
Chg2Cap [31]	72.04	60.18	**50.84**	**41.70**	28.92	58.66	92.49	55.44
Sparse Focus [56]	67.30	55.97	47.00	37.30	26.32	56.38	91.59	52.90
DiffusionRSCC [57]	-	-	-	33.30	27.40	56.50	88.70	51.48
SEN [58]	70.95	57.28	45.81	36.25	26.62	55.95	91.77	52.65
SCNet(ours)	**73.48**	**60.98**	50.51	40.80	**29.62**	**59.06**	**97.74**	**56.81**

TABLE III
COMPARISON WITH THE STATE OF THE ART ON THE WHU-CDC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	$S_m^\star$
Capt-Rep-Diff [37]	70.59	59.02	50.70	45.33	32.29	63.44	108.07	62.28
Capt-Att [37]	75.62	64.68	56.89	50.62	33.97	67.17	118.33	67.52
Capt-Dual-Att [37]	77.15	68.47	60.81	55.42	34.22	68.04	121.45	69.78
DUDA [37]	79.04	69.53	61.57	55.64	34.29	68.98	121.85	70.19
MCCFormers-S [55]	81.12	75.04	69.95	65.34	42.11	78.52	147.09	83.27
MCCFormers-D [55]	73.29	67.88	64.03	60.96	39.69	73.67	134.92	77.81
RSICCformer [26]	80.05	74.24	69.61	66.54	42.65	73.91	133.44	79.64
PSNet [32]	81.26	73.25	65.78	60.32	36.97	71.60	130.52	74.85
Prompt-CC [42]	81.12	73.96	67.22	61.45	36.99	71.88	134.50	76.21
Chg2Cap-S [31]	77.27	71.33	65.10	58.44	39.95	78.67	**160.64**	84.43
Chg2Cap [31]	78.93	72.64	67.20	62.71	41.46	77.95	144.18	81.58
Sparse Focus [56]	81.17	72.90	66.06	60.27	37.34	72.63	134.64	76.72
DiffusionRSCC [57]	75.32	70.15	66.40	63.76	40.18	73.80	127.96	76.43
Mask Approx Net [30]	81.34	75.68	71.16	67.73	43.89	75.41	135.31	80.59
SEN [58]	80.60	74.64	67.69	61.97	36.76	71.70	133.57	76.00
SCNet (ours)	**84.47**	**78.72**	**73.45**	**69.01**	**44.44**	**80.48**	153.54	**86.87**

3) *WHU-CDC Dataset (Table III):* To evaluate generalization, we tested on the WHU-CDC dataset. SCNet exhibits exceptional dominance, ranking first in **7 out of 8 metrics**. It achieves **69.01% in BLEU-4** and **86.87% in** $S_m^\star$, surpassing other methods by clear margins. While Chg2Cap-S shows a specific advantage in CIDEr, SCNet significantly outperforms the standard Chg2Cap (153.54% versus 144.18%) and secures the best results in all linguistic fluency metrics (BLEU-1–BLEU-4), ensuring high-quality caption generation.

Overall, the consistent top-tier performance across three diverse datasets confirms the effectiveness of our lightweight "Encoder-Heavy, Fusion-Light" design. By jointly modeling spatial and channel dependencies, SCNet effectively captures fine-grained changes and generalizes well across different domains.

D. Ablation Studies

We conduct a comprehensive series of experiments on the LEVIR-CC dataset to validate the effectiveness of the components proposed in our method. Additionally, we explore the impact of various factors on the results, including different dimensions for computing cosine similarity, different fusion methods, and different feature map spatial sizes. In the ablation studies, all experimental settings remain consistent with those described in the previous experiments.

1) Effectiveness of Different Components: Table IV shows the impact of the SEM and ConvFFN [67] modules on the model's performance. We can observe that the inclusion of SEM and ConvFFN significantly enhances the model's performance, highlighting the importance of these two modules. Specifically, compared to the method without SEM and ConvFFN, the simultaneous addition of these modules achieves an improvement of 1.69% in BLEU-4 and 2.17% in CIDEr-D, with only a minimal increase in parameters (0.05 M).

2) Impact of CosineSimilarity Along Different Dimensions: Inspired by Chg2Cap [31], we introduce a cosine similarity bias to enhance feature fusion. Since the optimal dimension for similarity computation has not been thoroughly

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TABLE IV

EFFECTIVENESS OF SEM AND CONVFFN ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

SEM	ConvFFN	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Param
✗	✗	85.37	76.97	69.96	64.13	40.33	74.77	138.06	20.35M
✓	✗	85.62	77.21	69.92	63.80	40.41	75.12	139.39	21.94M
✗	✓	85.70	77.54	70.40	64.70	39.90	74.82	136.88	18.82M
✓	✓	**86.14**	**78.19**	**71.44**	**65.82**	**40.51**	**75.37**	**140.23**	20.40M

TABLE V

ABLATION STUDY ON CALCULATING COSINE SIMILARITY ALONG DIFFERENT DIMENSIONS. "DIM" DENOTES THE DIMENSION(S) USED FOR CALCULATION. THE BEST RESULTS OF OUR SELECTED STRATEGY ARE HIGHLIGHTED IN BOLD

Calculation Dimension	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
Base (w/o Similarity)	84.97	76.65	69.64	63.85	39.71	74.01	136.07
Spatial (H)	86.09	77.90	70.96	65.23	40.45	75.41	139.87
Spatial (W)	86.02	77.81	70.89	65.26	40.39	75.21	139.80
Spatial ($H \times W$)	86.02	77.65	70.65	64.99	40.34	75.30	139.67
Spatial ($H + W$)	85.99	77.94	71.08	65.55	40.12	75.03	139.75
Channel (C)	**86.14**	**78.19**	**71.44**	**65.82**	**40.51**	**75.37**	**140.23**
Channel + Spatial ($C + H \times W$)	85.73	77.46	70.58	64.94	40.16	74.86	138.83
All Dims ($C + H + W$)	85.93	77.81	71.01	65.47	40.46	75.48	140.53

TABLE VI

IMPACT OF DIFFERENT FUSION METHODS IN DIFFERENCE-GUIDED FUSION MODULE ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Fusion methods	B1	B4	M	R	C
Sub	86.07	64.15	40.42	75.36	138.46
Sum	85.26	65.31	39.86	74.92	138.07
Element-wise Product	84.96	64.53	39.99	74.70	138.07
Concatenate	**86.14**	**65.82**	**40.51**	**75.37**	**140.23**

TABLE VII

IMPACT OF DIFFERENT FEATURE MAP SIZE ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Feature map Size	B1	B4	M	R	C
7×7	85.19	63.25	40.33	75.14	138.94
8×8	**86.14**	**65.82**	**40.51**	**75.37**	**140.23**
9×9	86.05	64.44	40.37	75.15	139.08

TABLE VIII

PERFORMANCE COMPARISONS WITH DIFFERENT DEPTHS OF ENCODER AND DECODER ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%) EXCEPT FOR PARAMETER, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

E	D	B1	B4	M	R	C	Param
1	1	85.56	64.65	39.29	74.39	135.18	13.01M
2	1	84.69	64.86	39.95	74.75	136.55	16.70M
3	1	**86.14**	**65.82**	**40.51**	**75.37**	**140.23**	20.40M
3	2	85.93	64.82	**40.51**	75.21	138.96	25.13M
3	3	85.51	64.30	40.09	74.91	138.37	29.86M

explored, we conduct an ablation study by calculating cosine similarity along different axes, including spatial (H, W), channel (C), and their combinations. Table V results show that introducing similarity bias along any dimension consistently improves performance over the baseline, validating its effectiveness. Among single-dimensional strategies, the channel-wise ($C \times C$) computation achieves the largest gain (140.23% CIDEr), outperforming spatial-based approaches, indicating the importance of semantic correlation for captioning. Although combining all dimensions ($C + H + W$) yields a slightly higher score (140.53%), the marginal improvement (+0.3%) comes with increased computational cost. Therefore, we adopt the channel-wise (C) strategy as the default setting for efficiency and effectiveness.

3) Impact of Different Fusion Methods in Difference-Guided Fusion: In Table VI, we compare four different methods for fusing bi-temporal image pair features. The experimental results indicate that concatenating the features of the two images yields the best performance. By concatenating the image features and then adding a 1×1 convolutional layer, we can rearrange the concatenated features to form a new feature, thereby minimizing the semantic information loss of the two features refined by the transformer encoder.

4) Impact of Different Feature Map Size: In Table VII, we validate the impact of different feature map spatial sizes on model performance. The feature maps extracted by ResNet-101 have a spatial size of 8×8. Subsequently, we use average pooling to set the spatial size to 7×7 and 9×9. The results indicate that the model already achieves optimal performance with the default spatial size of 8×8.

5) Analysis of Network Structure Complexity: In a transformer-based encoder–decoder architecture, the layer

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TABLE IX

IMPACT OF DIFFERENT LOCAL ENHANCE METHODS IN CAM ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	B1	B4	M	R	C
SE [52] (r=4)	85.45	63.56	40.38	**75.59**	139.95
SE [52] (r=8)	85.59	64.23	39.66	75.16	138.60
SE [52] (r=16)	85.94	.64.47	40.39	75.45	140.11
SE [52] (r=32)	85.62	63.64	39.09	74.24	134.09
1×1 Conv	86.06	64.66	40.47	75.23	138.56
3×3 Conv	85.72	64.30	40.30	75.13	138.84
DWConv	**86.14**	**65.82**	**40.51**	75.37	**140.23**

TABLE X

PERFORMANCE COMPARISON OF DIFFERENT COMBINATIONS OF SAM AND CAM ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Combinations	B1	B4	M	R	C
SAM	85.82	64.53	40.39	74.99	138.67
CAM	85.85	64.30	39.90	74.31	136.42
SAM → CAM	**86.14**	**65.82**	**40.51**	**75.37**	**140.23**
CAM → SAM	85.51	65.12	40.30	75.06	139.39
SAM + CAM	85.70	65.26	40.22	75.14	139.10

count of both the encoder and decoder is a crucial hyperparameter. It significantly impacts the model's complexity, computational overhead, and overall effectiveness. To comprehensively assess the optimal configuration of encoder and decoder layers, Table VIII presents a detailed performance comparison of various combinations. It can be observed that as the number of transformer encoder layers increases, the model's performance gradually improves. However, with the same number of encoder layers, the model's performance decreases as the number of decoder layers increases. This decline in performance may be due to the increased difficulty in optimizing the model with more decoder layers, making it more challenging to generate high-quality captions. Therefore, we finally selected a model architecture consisting of three encoder layers and a single decoder layer. This configuration strikes an optimal balance between model capacity and training efficiency, enabling effective feature extraction while minimizing the risk of overfitting and ensuring efficient caption generation.

6) Impact of Different Local Enhance Methods in CAM: To extract both global and local information from the images, we introduce a local enhancement branch in the CAM. In Table IX, we compare different local enhancement methods. Compared to other methods, incorporating a 3×3 DWConv achieved the best results across most metrics.

7) Effect of Different Combinations of Spatial-Attention Module and CAM: Since SAM and CAM model the spatial and channel aspects, respectively, their combination sequence significantly affects the model's performance. Therefore, in this part, we primarily investigate three different combination methods of SAM and CAM: first passing through SAM then CAM (SAM → CAM), first passing through CAM then SAM (CAM → SAM), and passing through SAM and CAM in parallel (SAM + CAM).

Table X lists the performance of the three different combination methods. To validate the importance of joint spatial and channel modeling, the performance of using only SAM or CAM is also included. Compared to modeling only the spatial or channel aspects, all three combination methods that integrate both show improved performance. This indicates that both spatial attention and channel attention are crucial for the change captioning task. Moreover, among the three combination methods, the approach of first passing through SAM and then through CAM performs the best.

TABLE XI

COMPARISON OF MODEL PARAMETERS AND COMPUTATIONAL COST (FLOPS) ACROSS DIFFERENT METHODS WITH 256×256 IMAGE SIZE. NOTE THAT ALL REPORTED VALUES CORRESPOND TO THE TASK-SPECIFIC MODULES, EXCLUDING THE PRETRAINED BACKBONE PARAMETERS AND COMPUTATIONS

ID	Method	Paramters	FLOPs
1	Capt-Rep-Diff	45.6M	19.8G
2	Capt-Att	46.1M	19.9G
3	Capt-Dual-Att	48.1M	20.0G
4	DUDA	52.8M	20.3G
5	MCCFormer-S	135.0M	25.1G
6	MCCFormer-D	135.0M	25.1G
7	RSICCFormer	145.3M	27.1G
8	PSNet	231.5M	13.8G
9	Prompt-CC	196.3M	19.9G
10	Chg2Cap	286.5M	20.2G
11	Sparse Focus	117.5M	6.5G
12	SEN	39.9M	10.9G
13	DiffusionRSCC	103.0M	15.2T
14	Mask Apporx Net	202.47M	12.3T
15	SCNet(ours)	20.40M	2.1G

8) Analysis of Parameters and FLOPs and FPS: Table XI presents a comparative analysis of task-specific modules (excluding backbones) across different methods. Notably, SCNet achieves the most compact architecture with 20.40 M parameters, representing a massive reduction to approximately 1/14th the size of Chg2Cap (286.5 M).

In terms of computational complexity, SCNet demonstrates superior efficiency with only 2.1 G FLOPs. This value is drastically lower than most methods and surpasses even lightweight competitors like SEN (10.9 G). This confirms that our "Encoder-Heavy, Fusion-Light" design successfully minimizes redundancy while maintaining high performance, positioning SCNet as a robust solution for resource-constrained applications.

Table XII compares the inference speed (FPS) of SCNet against other methods. While many models, for example, MCCFormer, operate slowly around 10–15 FPS, and lightweight methods like Sparse Focus reach around 32 FPS, Ours (SCNet) achieves the highest speed at 48.41 FPS. This result is approximately 50% faster than the runner-up Sparse Focus, confirming SCNet's superior efficiency and real-time capability for remote sensing image captioning tasks.

9) Effect of Layernorm Weight Sharing in SCE: In Table XIII, welist the impact of using shared layernorm on

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TABLE XII

COMPARISON OF REAL-TIME INFERENCE SPEED (FPS) FOR VARIOUS REMOTE SENSING CHANGE CAPTIONING METHODS

ID	Method	FPS
1	DUDA	33.14
2	MCCFormer-S	12.90
3	MCCFormer-D	12.40
4	RSICCformer	15.00
5	Prompt-CC	9.95
6	Chg2Cap	20.37
7	Sparse Focus	32.25
8	SEN	23.63
9	SCNet (ours)	48.41

TABLE XIII

EFFECT OF LAYER NORM WEIGHT SHARING IN SCE ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Combinations	B1	B4	M	R	C
CAM → SAM (w/o)	**85.55**	64.88	40.11	74.93	138.74
CAM → SAM (w/)	85.51	**65.12**	**40.30**	**75.06**	**139.39**
SAM → CAM (w/o)	86.03	65.06	40.43	75.30	139.91
SAM → CAM (w/)	**86.14**	**65.82**	**40.51**	**75.37**	**140.23**

model performance in two different combination scenarios. The results indicate that employing shared layernorm for SAM and CAM in the SCE under different conditions can effectively enhance model performance while reducing certain model parameters.

E. Further Comparisons

Following [26], we also employ the same three different aspects to evaluate the effectiveness of our method. These three aspects are: only test bi-temporal image pairs with no changes, only test bi-temporal image pairs with changes, and test the entire test set. The final results are reported in Table XIV.

It can be observed that for unchanged remote sensing images, the performance of Chg2Cap surpasses our method. This may be attributed to the introduction of SEM and CAM, making the model more sensitive to changes in small objects. Due to the resolution of the LEVIR-CC dataset being only 256 × 256, this can lead to some misjudgments by the model. However, it is precisely due to the inclusion of these two modules that our method significantly outperforms RSICCFormer and Chg2Cap in describing changed remote sensing image pairs. This further validates the importance of joint spatial–channel modeling. It not only captures changes in different regions of image pairs but also captures changes in different objects, thereby enhancing the model's ability to model image pairs and generate a more accurate image feature.

F. Visualization

In Figs. 5 and 6, we visualize two different examples from the LEVIR-CC dataset and the DUBAI-CCD dataset, respectively, to compare the ground truth captions, the captions generated by Chg2Cap [31], and the captions generated by our method.

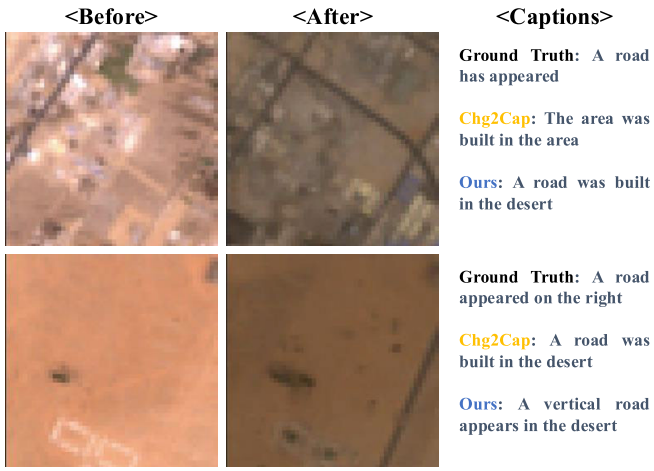


Fig. 5. Captioning results on the DUBAI-CCD dataset. Black sentence is one of the five GT sentences. The orange captions are generated by Chg2Cap [31], while the captions generated by our method are shown in blue.

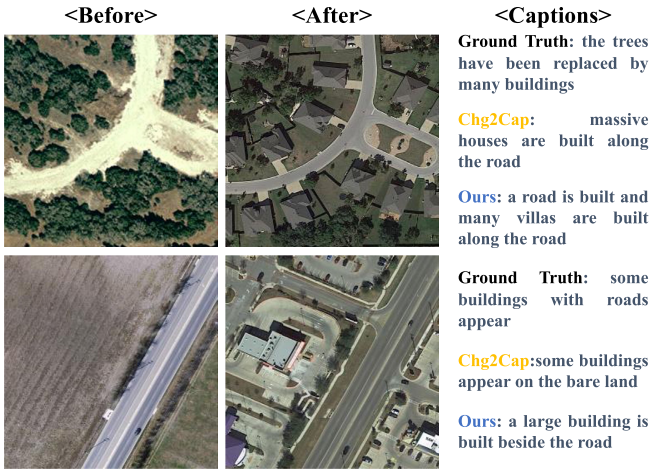


Fig. 6. Captioning results on the LEVIR-CC dataset. Black sentence is one of the five GT sentences. The orange captions are generated by Chg2Cap [31], while the captions generated by our method are shown in blue.

In the first example shown in Fig. 5, the caption generated by Chg2Cap is entirely unusable, whereas our method successfully identifies the construction of a road in the area and accurately determines that the background of the image pair is a desert environment. This demonstrates that by employing joint spatial–channel modeling, our model effectively focuses on the regions of change and captures the semantic information and attributes of the image pair. Furthermore, in the second example, our model not only identifies the road and the desert but also recognizes that the road is vertical, indicating that the model has learned richer semantic information and attributes of the road.

Furthermore, we can observe from Fig. 6 that in the first bi-temporal image pair, Chg2Cap successfully expressed that many houses were built along the road, but it lacked a description of the road itself. Our method not only depicted that villas were built along the road, but also described that the road was newly built. In the second bi-temporal image pair,

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TABLE XIV
PERFORMANCE COMPARISON UNDER THREE DIFFERENT ASPECTS ON THE LEVIR-CC DATASET. ALL VALUES ARE REPORTED AS PERCENTAGE (%), AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Test Set	Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
only no-change	RSICCFormer [26]	95.05	94.24	93.76	93.42	72.20	95.68	-
	Chg2Cap [31]	**98.08**	**97.83**	**97.70**	**97.61**	**78.21**	**98.22**	-
	Ours	97.80	97.44	97.33	97.29	76.23	97.49	-
only change	RSICCFormer [26]	76.43	61.92	48.81	38.14	25.72	52.53	60.56
	Chg2Cap [31]	**77.32**	**63.26**	50.06	39.09	25.73	52.02	58.30
	Ours	77.12	63.20	**51.20**	**41.46**	**26.28**	**53.23**	**68.91**
entire set	RSICCFormer [26]	84.72	76.27	68.87	62.77	39.61	74.12	134.12
	Chg2Cap [31]	**86.14**	78.04	70.66	64.39	40.03	75.12	136.61
	Ours	**86.14**	**78.19**	**71.44**	**65.82**	**40.51**	**75.37**	**140.23**

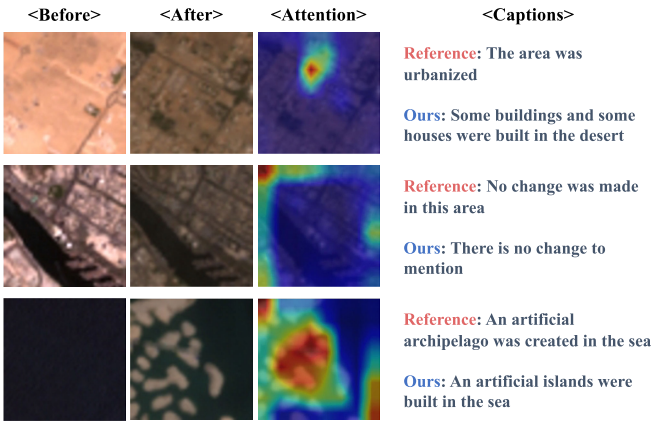


Fig. 7. Typical captioning results on the DUBAI-CCD dataset. (From left to right) prechange image, post-change image, attention map over post-change image, and captions from GT and generated by our method.

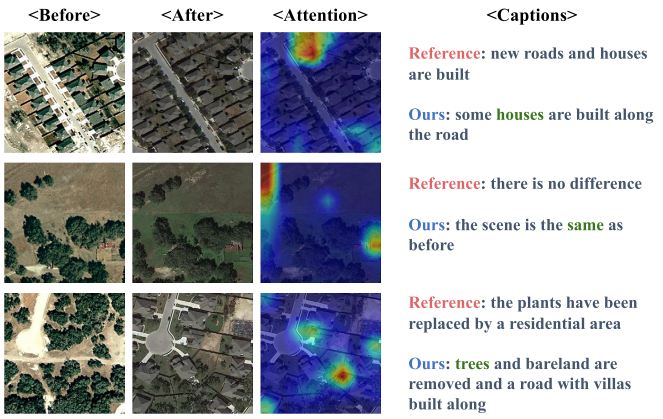


Fig. 8. Typical captioning results on the LEVIR-CC dataset. (From left to right) prechange image, post-change image, attention map over post-change image, and captions from GT and generated by our method.

the caption generated by our method not only successfully described that the building was built by the roadside but also emphasized that it is a large building. This indicates that after joint spatial–channel modeling, our method effectively focuses on different regions of the image while also paying attention to the features and attributes of different objects in the image. This further validates the effectiveness of our approach.

In Figs. 7 and 8, we visualize three examples from the LEVIR-CC dataset and the DUBAI-CCD dataset, respectively, to further compare the reference captions and the captions generated by our method. In these figures, "Before" denotes the image before the change, "After" denotes the image after the change, "Attention" represents the attention map over the image in the transformer decoder, "Reference" in the captions refers to the ground truth (GT), and "Ours" refers to the captions generated by our method.

In the DUBAI-CCD dataset, the spatial resolution of the change image pairs is 30 m, making many changes difficult to identify, particularly for features such as roads and houses. However, as shown in the first example of Fig. 7, the caption generated by our method, compared to the reference, not only depicts the urbanization of the area but also provides a more detailed description by identifying many buildings and houses being constructed in the desert. This demonstrates that our joint spatial–channel approach can recognize the desert background of the image and also identify small objects like buildings and houses, as further validated by the attention map. In the second example, our caption accurately identifies that there is no change in the image pair. In the last example, the attention map shows that our model successfully focuses on the island part of the image and accurately describes the changes in that area. The examples from the LEVIR-CC dataset, as shown in Fig. 8, also demonstrate that our model can accurately focus its attention on specific target objects, such as "house" and "trees."

To better qualitatively evaluate the captions generated by our method, we visualized each generated word and its corresponding attended images in Figs. 9 and 10. Notably, in Fig. 9, compared to the word "road," the attended image for the word "desert" broadens the scope of attention, thereby better identifying the background category of the image. These examples not only verify that our model can accurately recognize changes in different regions of the image but also highlight its ability to capture the semantic information of various objects in the image, resulting in more detailed descriptions of the image pairs. This demonstrates that joint spatial–channel modeling can significantly improve the accuracy and detail of

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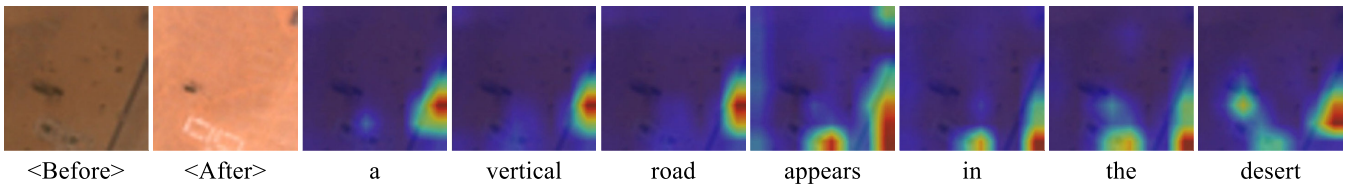


Fig. 9. Visualization of the step-by-step caption generation process on the DUBAI-CCD dataset. The heatmaps highlight the specific spatial regions SCNet focuses on when generating each word. This demonstrates the model's ability to accurately align visual changes (e.g., "road" and "desert") with linguistic descriptions.

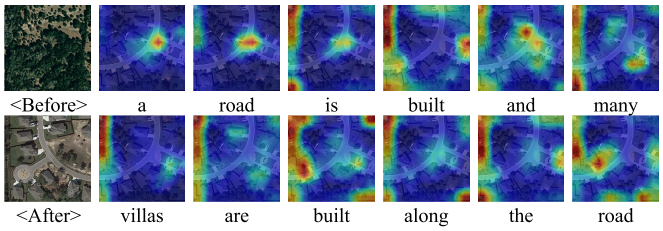


Fig. 10. Visualization of attention maps during caption generation on the LEVIR-CC dataset. The sequence displays the attention weights associated with each generated word. Brighter regions indicate higher attention, verifying that SCNet effectively captures and distinguishes fine-grained details (e.g., "villas" and "road") within the complex background.

captions generated for the RSICC task. From Fig.10, it can be observed that our proposed methode ectively focuses attention on objects such as roads and villas, enabling the generation of more detailed descriptions.

V. Conclusion

In this article, we propose a novel transformer-based method SCNet for the RSICC task that utilizes joint spatial–channel modeling. The proposed SCE module effectively addresses the challenges of modeling spatial and channel dependencies in remote sensing images. By sequentially capturing relationships between image regions and the semantic attributes of objects, the SCE enhances feature representation and enables accurate generation of change captions. The integration of depth-wise convolution improves local information modeling, while shared encoder parameters optimize computational efficiency for bi-temporal image pairs. Unlike traditional multistage fusion approaches, this method simplifies feature fusion through concatenation and a residual block, achieving precise descriptions of change regions and object transformations.

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